Forcelink Partner Integration API

Momentum Dashboard POC - Developer Guide

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This guide explains how to call the Forcelink partner API that feeds the Momentum dashboards: time & attendance, proof of service, stock, resources, escalations, Prestige Pulse QR-code customer feedback, and cleaner tracking (NFC scans and beacons). For each endpoint it gives the parameters, every response field and a real example response, then shows how to combine the endpoints into the dashboard measures.

The example data comes from the mock API, which serves a year of generated data (17 Sep 2025 - 17 Sep 2026) in exactly the production response format.

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1. Getting started

Every endpoint is a read-only **GET** that returns JSON. You need a base URL and an API key.

Environment	Base URL
Mock (local)	`http://127.0.0.1:8000`
Mock (Docker)	`http://localhost:8000` after `docker compose up --build`
SwaggerHub mock	`https://virtserver.swaggerhub.com/acumensoftware/forcelink-partner-integration-api/1`

Interactive documentation (Swagger UI) is at `<base URL>/docs` on the local mock. To run the mock from the repository:

```
./run.sh                                # venv, install, seed, serve
# or
python3 -m venv .venv && .venv/bin/pip install -r requirements.txt
.venv/bin/python -m app.seed --reset
.venv/bin/uvicorn app.main:app --port 8000
```

A first request - the examples in this guide assume these two shell variables:

```
export BASE_URL="http://127.0.0.1:8000"
export FORCELINK_API_KEY="fl_uat_momentum_7f3c9a21d4e84b0c95e6"   # mock key

curl -H "X-Api-Key: $FORCELINK_API_KEY" \
  "$BASE_URL/resources?orgUnitId=4021&per_page=5"
```

2. Authentication and scoping

Send the API key in the `X-Api-Key` header on every request. `Authorization: Bearer <key>` is also accepted. A missing or wrong key returns **401 UNAUTHORIZED**.

Each key is scoped to a set of sites (org units) and stores. Responses only ever contain data for the key's sites. Asking explicitly for a site or store outside that scope returns **403 FORBIDDEN**.

Scope (mock key)	Values
Sites (orgUnitId)	4021 Head Office - Centurion, 4022 Parktown Campus, 4023 Durbanville Office, 4024 Umhlanga Ridge Office, 4025 Menlyn Office
Stores (storeId)	12, 13, 14, 15
Out of scope (for testing 403)	site 4099, store 99

A real UAT key, scoped to the agreed Momentum sites, is issued separately. Keep it server-side - never embed it in browser code.

3. Conventions

3.1 Response envelope

Every response, success or failure, uses the same envelope:

Field	Meaning
`success`	true on success, false on error.
`data`	Array of records (null on error).
`message`	Human-readable message on error, otherwise null.
`error`	Error object: `code`, `message`, optional `details` array (per-field problems).
`pagination`	`page`, `per_page`, `total_items`, `total_pages`.

Field	Meaning
`meta`	`request_id`, `version`, `timestamp`. The request id is also returned in the `X-Request-Id` header - quote it when reporting a problem.

3.2 Pagination

All endpoints are paged with `page` (1-based, default 1) and `per_page` (default 50, maximum 500). Keep requesting pages until `page` reaches `pagination.total_pages`:

```
import requests

def fetch_all(path, **params):
    params = {**params, "per_page": 500, "page": 1}
    headers = {"X-Api-Key": API_KEY}
    while True:
        body = requests.get(BASE_URL + path, params=params, headers=headers).json()
        if not body["success"]:
            raise RuntimeError(body["error"])
        yield from body["data"]
        if params["page"] >= body["pagination"]["total_pages"]:
            break
        params["page"] += 1

rows = list(fetch_all("/customer-feedback/qr-responses",
                      dateFrom="2026-09-01", dateTo="2026-09-17"))
```

3.3 Dates and timestamps

- `dateFrom` and `dateTo` are ISO-8601 dates (`YYYY-MM-DD`), both inclusive. `dateTo` may not be before `dateFrom`, and one request may span at most **92 days**. For longer periods, split the period into chunks and combine the results.
- Each endpoint filters on one timestamp, given in its reference section (for example, escalations filter on `createdDate`).
- All timestamps are UTC, in the form `2026-09-01T07:05:00Z`. Convert to SAST (UTC+2) for display.

3.4 Errors

HTTP	error.code	When
400	`VALIDATION_ERROR`	Missing or malformed parameter, bad date range, unknown enum value (e.g. `status=FOO`), paging out of range.
401	`UNAUTHORIZED`	Missing or invalid API key.
403	`FORBIDDEN`	Site or store outside the key's scope.
429	`RATE_LIMITED`	Too many requests - back off and retry.
500	`INTERNAL_ERROR`	Unexpected server error - retry, then report with the request id.

Example error response (`dateTo` before `dateFrom`):

```
{
  "success": false,
  "data": null,
  "message": "dateTo must be after dateFrom.",
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "dateTo must be after dateFrom.",
    "details": [
      {
        "field": "dateTo",
        "code": "must_be_after",
        "message": "dateTo must be after dateFrom."
      }
    ]
  },
  "pagination": null,
  "meta": {
    "request_id": "req_MYC4K9CWYEPXTSQFG4JT5914QD",
    "version": "1",
    "timestamp": "2026-10-29T12:24:03Z"
  }
}
```

4. Endpoint reference

Endpoint	Returns
`/attendance/shift-records`	Per-shift attendance records for a date range.
`/service-proof/completed-work-orders`	Completed work orders for a date range, by site.
`/stock/on-hand`	Current on-hand stock quantities, by store.
`/stock/movements`	Stock movements/transactions for a date range, by store.
`/resources`	Resource (employee / user) lookup, including supervisor hierarchy.
`/escalations`	Escalations raised in a date range, by site.
`/customer-feedback/qr-responses`	QR-code customer feedback submissions for a date range, by site.
`/cleaner-tracking/check-ins`	Cleaner NFC tag scans and beacon check-ins for a date range, by site.

4.1 Per-shift attendance records for a date range

`GET /attendance/shift-records`

Returns one record per employee per shift/day within the requested range: clock in/out timestamps, a derived late/absent flag, and leave status. Grouped by `employeeId` + `orgUnit` (site) as agreed with Momentum. No real-time/live status - this is a per-shift/day batch view.

- One row per employee per shift day. **employeeId** is the same id as **resourceId** in /resources.
- Clock in/out are null when the employee was ABSENT or ON_LEAVE.

Query parameters

Parameter	Type	Required	Description
`dateFrom`	date	Yes	Inclusive start of the date range (ISO-8601 date, `YYYY-MM-DD`). Example: `2026-09-01`.
`dateTo`	date	Yes	Inclusive end of the date range (ISO-8601 date, `YYYY-MM-DD`). Maximum span per request is 92 days. Example: `2026-09-30`.
`orgUnitId`	integer	No	Filter to a single site. `orgUnit.type` is `SITE` for every value this endpoint accepts; omit to return all sites the API key is scoped to. Example: `4021`.
`employeeId`	integer	No	Filter to a single employee/resource. Omit to return all employees at the requested site(s). Example: `88213`.
`page`	integer	No	1-indexed page number. Defaults to the first page.
`per_page`	integer	No	Items per page. Default 50, max 500.

Response fields (each item in `data`)

Field	Type	Notes
`employeeId`	integer	Forcelink resource/employee id.
`employeeCode`	string	Employee's payroll/staff code, if configured.
`orgUnit`	object	The site the shift is planned against.
`orgUnit.id`	integer	
`orgUnit.type`	string	One of: `SITE`
`orgUnit.name`	string	e.g. `Head Office - Centurion`
`shiftDate`	date	Calendar date the shift falls on (per-shift/day granularity, not a live feed).
`clockInTimestamp`	date-time, nullable	Null if the employee never clocked in for this shift.
`clockOutTimestamp`	date-time, nullable	
`status`	string	Derived attendance flag for the shift. One of: `ON_TIME`, `LATE`, `ABSENT`, `ON_LEAVE`
`leaveStatus`	string	`NONE` unless a leave request overlaps this date. One of: `NONE`, `PENDING`, `APPROVED`, `REJECTED`

Example request

```
curl -G "$BASE_URL/attendance/shift-records" \
  -H "X-Api-Key: $FORCELINK_API_KEY" \
  -d dateFrom=2026-09-01 -d dateTo=2026-09-15 -d orgUnitId=4021 -d per_page=1
```

Example response (first item; full envelope in section 3.1)


```
{
  "success": true,
  "data": [
    {
      "employeeId": 88201,
      "employeeCode": "EMP88201",
      "orgUnit": {
        "id": 4021,
        "type": "SITE",
        "name": "Head Office - Centurion"
      },
      "shiftDate": "2026-09-01",
      "clockInTimestamp": "2026-09-01T06:50:00Z",
      "clockOutTimestamp": "2026-09-01T16:56:00Z",
      "status": "ON_TIME",
      "leaveStatus": "NONE"
    }
  ],
  "pagination": {
    "page": 1,
    "per_page": 1,
    "total_items": 242,
    "total_pages": 242
  },
  "meta": "..."
}
```

4.2 Completed work orders for a date range, by site

`GET /service-proof/completed-work-orders`

Returns work orders with status COMPLETED within the requested date range, grouped by site. Serves as proof-of-service for the POC. SLA status / breach-flag fields are intentionally not included - that calculation is out of scope for this exercise and will be spec'd at a later stage.

- Only COMPLETED work orders are returned; the date range filters on **completedDate**.
- **createdDate** is new: completion time = completedDate – createdDate.

Query parameters

Parameter	Type	Required	Description
`dateFrom`	date	Yes	Inclusive start of the date range (ISO-8601 date, `YYYY-MM-DD`). Example: `2026-09-01`.
`dateTo`	date	Yes	Inclusive end of the date range (ISO-8601 date, `YYYY-MM-DD`). Maximum span per request is 92 days. Example: `2026-09-30`.
`orgUnitId`	integer	No	Filter to a single site. `orgUnit.type` is `SITE` for every value this endpoint accepts; omit to return all sites the API key is scoped to. Example: `4021`.
`page`	integer	No	1-indexed page number. Defaults to the first page.
`per_page`	integer	No	Items per page. Default 50, max 500.

Response fields (each item in `data`)

Field	Type	Notes
`workOrderId`	integer	
`workOrderCode`	string	e.g. `WO-2026-104822`
`orgUnit`	object	
`orgUnit.id`	integer	
`orgUnit.type`	string	One of: `SITE`
`orgUnit.name`	string	
`status`	string	Fixed to COMPLETED for this exercise; SLA-status and breach-flag fields are out of scope for the POC and will be spec'd separately. One of: `COMPLETED`
`workOrderType`	string	e.g. `Scheduled Clean`
`description`	string	
`createdDate`	date-time	When the work order was created/logged. Completion time = `completedDate - createdDate`.
`completedDate`	date-time	
`resourceId`	integer, nullable	Resource (cleaner) who closed the work order, if assigned. Resolve via `/resources`.

Example request

```
curl -G "$BASE_URL/service-proof/completed-work-orders" \
  -H "X-Api-Key: $FORCELINK_API_KEY" \
  -d dateFrom=2026-09-01 -d dateTo=2026-09-17 -d orgUnitId=4021 -d per_page=1
```

Example response (first item; full envelope in section 3.1)


```
{
  "success": true,
  "data": [
    {
      "workOrderId": 117192,
      "workOrderCode": "WO-2026-117192",
      "orgUnit": {
        "id": 4021,
        "type": "SITE",
        "name": "Head Office - Centurion"
      },
      "status": "COMPLETED",
      "workOrderType": "Waste Removal",
      "description": "Removed recycling to central refuse area",
      "createdDate": "2026-09-01T06:05:00Z",
      "completedDate": "2026-09-01T09:30:00Z",
      "resourceId": 88217
    }
  ],
  "pagination": {
    "page": 1,
    "per_page": 1,
    "total_items": 127,
    "total_pages": 127
  },
  "meta": "..."
}
```

4.3 Current on-hand stock quantities, by store

`GET /stock/on-hand`

Returns current on-hand quantity per material, grouped by store.

- Current snapshot, not date-ranged. Filter by **storeId** (stores 12–15 are in scope).

Query parameters

Parameter	Type	Required	Description
`storeId`	integer	No	Filter to a single store. Omit to return all stores the API key is scoped to. Example: 12.
`page`	integer	No	1-indexed page number. Defaults to the first page.
`per_page`	integer	No	Items per page. Default 50, max 500.

Response fields (each item in `data`)

Field	Type	Notes
`store`	object	
`store.id`	integer	
`store.name`	string	e.g. `Cleaning Store - Centurion`
`materialId`	integer	
`materialCode`	string	e.g. `MAT-00231`
`materialDescription`	string	e.g. `Toilet Paper, 2-Ply 48-Roll Pack`
`quantityOnHand`	number	
`unitOfMeasure`	string	e.g. EA
`asOfTimestamp`	date-time	When this on-hand snapshot was taken.

Example request

```
curl -G "$BASE_URL/stock/on-hand" \
  -H "X-Api-Key: $FORCELINK_API_KEY" \
  -d storeId=12 -d per_page=1
```

Example response (first item; full envelope in section 3.1)

```
{
  "success": true,
  "data": [
    {
      "store": {
        "id": 12,
        "name": "Cleaning Store - Centurion"
      },
      "materialId": 9001,
      "materialCode": "MAT-00231",
      "materialDescription": "Toilet Paper, 2-Ply 48-Roll Pack",
      "quantityOnHand": 417.0,
      "unitOfMeasure": "PACK",
      "asOfTimestamp": "2026-09-17T06:00:00Z"
    }
  ],
  "pagination": {
    "page": 1,
    "per_page": 1,
    "total_items": 24,
    "total_pages": 24
  },
  "meta": "..."
}
```

4.4 Stock movements/transactions for a date range, by store

`GET /stock/movements`

Returns stock movement/transaction history (receipts, issues, transfers, adjustments), grouped by store. Supports both current and historical queries - pass any `dateFrom`/`dateTo` range.

- Positive quantities are inflows (RECEIPT, TRANSFER_IN), negative are outflows (ISSUE, TRANSFER_OUT); ADJUSTMENT can be either.
- ISSUE rows reference the work order the stock was used on.

Query parameters

Parameter	Type	Required	Description
`dateFrom`	date	Yes	Inclusive start of the date range (ISO-8601 date, `YYYY-MM-DD`). Example: `2026-09-01`.
`dateTo`	date	Yes	Inclusive end of the date range (ISO-8601 date, `YYYY-MM-DD`). Maximum span per request is 92 days. Example: `2026-09-30`.
`storeId`	integer	No	Filter to a single store. Omit to return all stores the API key is scoped to. Example: `12`.
`page`	integer	No	1-indexed page number. Defaults to the first page.
`per_page`	integer	No	Items per page. Default 50, max 500.

Response fields (each item in `data`)

Field	Type	Notes
`movementId`	integer	
`store`	object	
`store.id`	integer	
`store.name`	string	
`materialId`	integer	
`materialCode`	string	
`materialDescription`	string	
`transactionType`	string	One of: `RECEIPT`, `ISSUE`, `TRANSFER_IN`, `TRANSFER_OUT`, `ADJUSTMENT`
`quantity`	number	Signed where relevant (e.g. negative for ISSUE).
`transactionTimestamp`	date-time	
`referenceDocument`	string, nullable	Linked work order code or purchase-order/GRN number, where applicable.

Example request

```
curl -G "$BASE_URL/stock/movements" \
  -H "X-Api-Key: $FORCELINK_API_KEY" \
  -d dateFrom=2026-09-01 -d dateTo=2026-09-17 -d storeId=12 -d per_page=1
```

Example response (first item; full envelope in section 3.1)


```
{
  "success": true,
  "data": [
    {
      "movementId": 512026,
      "store": {
        "id": 12,
        "name": "Cleaning Store - Centurion"
      },
      "materialId": 9017,
      "materialCode": "MAT-01110",
      "materialDescription": "Dishwashing Liquid, 5L",
      "transactionType": "TRANSFER_OUT",
      "quantity": -17.0,
      "transactionTimestamp": "2026-09-01T06:05:00Z",
      "referenceDocument": "TRF-2026-2760"
    }
  ],
  "pagination": {
    "page": 1,
    "per_page": 1,
    "total_items": 156,
    "total_pages": 156
  },
  "meta": "..."
}
```

4.5 Resource (employee / user) lookup, including supervisor hierarchy

`GET /resources`

Returns the resources (cleaners, supervisors, site managers) at the sites the API key is scoped to. Use it to resolve the resource ids returned by the other endpoints, and `supervisorId` to roll cleaner metrics up to their supervisor. Not date-ranged.

- Not date-ranged. Use it as a lookup table: fetch once per site and cache.
- **supervisorId** gives the reporting line: cleaner → supervisor → site manager.

Query parameters

Parameter	Type	Required	Description
`orgUnitId`	integer	No	Filter to a single site; omit to return all sites the API key is scoped to. Example: `4021`.
`resourceId`	integer	No	Return a single resource. Example: `88213`.
`role`	string	No	Filter by role. One of: `CLEANER`, `SUPERVISOR`, `SITE_MANAGER`. Example: `SUPERVISOR`.
`page`	integer	No	1-indexed page number. Defaults to the first page.
`per_page`	integer	No	Items per page. Default 50, max 500.

Response fields (each item in `data`)

Field	Type	Notes
`resourceId`	integer	Forcelink resource id. Same id space as `employeeId` (attendance), `resourceId` (work orders, check-ins) and the escalation resource fields.
`resourceCode`	string	e.g. `EMP88213`
`name`	string	e.g. `Nomsa Dlamini`
`role`	string	One of: `CLEANER`, `SUPERVISOR`, `SITE_MANAGER`
`orgUnit`	object	
`orgUnit.id`	integer	
`orgUnit.type`	string	One of: `SITE`
`orgUnit.name`	string	e.g. `Head Office - Centurion`
`supervisorId`	integer, nullable	Resource id of this resource's supervisor (cleaner -> supervisor, supervisor -> site manager). Null for site managers.
`active`	boolean	False once the resource has left / been deactivated.

Example request

```
curl -G "$BASE_URL/resources" \
  -H "X-Api-Key: $FORCELINK_API_KEY" \
  -d orgUnitId=4021 -d role=SUPERVISOR -d per_page=1
```

Example response (first item; full envelope in section 3.1)


```
{
  "success": true,
  "data": [
    {
      "resourceId": 88202,
      "resourceCode": "EMP88202",
      "name": "Sipho Coetzee",
      "role": "SUPERVISOR",
      "orgUnit": {
        "id": 4021,
        "type": "SITE",
        "name": "Head Office - Centurion"
      },
      "supervisorId": 88201,
      "active": true
    }
  ],
  "pagination": {
    "page": 1,
    "per_page": 1,
    "total_items": 3,
    "total_pages": 3
  },
  "meta": "..."
}
```

4.6 Escalations raised in a date range, by site

`GET /escalations`

Returns escalations whose `createdDate` falls in the requested range. Response times are derived from `createdDate`, `acknowledgedDate` and `resolvedDate`.

- The date range filters on **createdDate**. An escalation that was raised before the range but resolved inside it is not returned - widen the range if you need it.
- **level** rises to 2 once open for more than 4h and to 3 (assigned to the site manager) once open for more than 24h.

Query parameters

Parameter	Type	Required	Description
`dateFrom`	date	Yes	Inclusive start of the date range (ISO-8601 date, `YYYY-MM-DD`). Example: `2026-09-01`.
`dateTo`	date	Yes	Inclusive end of the date range (ISO-8601 date, `YYYY-MM-DD`). Maximum span per request is 92 days. Example: `2026-09-30`.
`orgUnitId`	integer	No	Filter to a single site; omit to return all sites the API key is scoped to. Example: `4021`.
`status`	string	No	Filter by current status. One of: `OPEN`, `ACKNOWLEDGED`, `RESOLVED`. Example: `OPEN`.
`page`	integer	No	1-indexed page number. Defaults to the first page.
`per_page`	integer	No	Items per page. Default 50, max 500.

Response fields (each item in `data`)

Field	Type	Notes
`escalationId`	integer	
`escalationCode`	string	e.g. `ESC-2026-606081`
`orgUnit`	object	
`orgUnit.id`	integer	
`orgUnit.type`	string	One of: `SITE`
`orgUnit.name`	string	e.g. `Head Office - Centurion`
`location`	object, nullable	Area within the site where the QR code / NFC tag / beacon is installed.
`location.id`	integer	
`location.name`	string	e.g. `Level 2 - Ladies Restroom`
`source`	string	One of: `QR_FEEDBACK`, `SUPERVISOR`, `CLIENT_REQUEST`
`feedbackId`	integer, nullable	QR feedback submission that triggered this escalation (source `QR_FEEDBACK` only).
`workOrderId`	integer, nullable	Reactive work order raised to action the escalation, if any.
`category`	string	One of: `CLEANLINESS`, `CONSUMABLES`, `ODOUR`, `SPILL`, `BROKEN_FIXTURE`, `GENERAL`
`priority`	string	One of: `LOW`, `MEDIUM`, `HIGH`, `CRITICAL`
`level`	integer	Escalation level: 1 = supervisor, 2 = supervisor (escalated, open > 4h), 3 = site manager (open > 24h). Range 1–3.
`status`	string	One of: `OPEN`, `ACKNOWLEDGED`, `RESOLVED`
`raisedByResourceId`	integer, nullable	Resource who raised it (source `SUPERVISOR`); null for customer / client-raised escalations.
`assignedToResourceId`	integer, nullable	Supervisor or site manager the escalation is assigned to.
`createdDate`	date-time	When the escalation was raised.

Field	Type	Notes
`acknowledgedDate`	date-time, nullable	When the assignee acknowledged it.
`resolvedDate`	date-time, nullable	When it was resolved.

Example request

```
curl -G "$BASE_URL/escalations" \
  -H "X-Api-Key: $FORCELINK_API_KEY" \
  -d dateFrom=2026-09-01 -d dateTo=2026-09-17 -d orgUnitId=4021 -d status=RESOLVED -d per_page=1
```

Example response (first item; full envelope in section 3.1)


```
{
  "success": true,
  "data": [
    {
      "escalationId": 605799,
      "escalationCode": "ESC-2026-605799",
      "orgUnit": {
        "id": 4021,
        "type": "SITE",
        "name": "Head Office - Centurion"
      },
      "location": {
        "id": 7003,
        "name": "Ground Floor - Accessible Restroom"
      },
      "source": "QR_FEEDBACK",
      "feedbackId": 352968,
      "workOrderId": null,
      "category": "CLEANLINESS",
      "priority": "HIGH",
      "level": 1,
      "status": "RESOLVED",
      "raisedByResourceId": null,
      "assignedToResourceId": 88204,
      "createdDate": "2026-09-01T07:51:00Z",
      "acknowledgedDate": "2026-09-01T08:47:00Z",
      "resolvedDate": "2026-09-01T09:32:00Z"
    }
  ],
  "pagination": {
    "page": 1,
    "per_page": 1,
    "total_items": 47,
    "total_pages": 47
  },
  "meta": "..."
}
```

4.7 QR-code customer feedback submissions for a date range, by site

`GET /customer-feedback/qr-responses`

Returns individual Prestige Pulse QR-code submissions whose `submittedTimestamp` falls in the requested range. Low ratings may have an associated escalation (`escalationId`).

- One row per QR-code submission (Prestige Pulse). Each QR code is fixed to a **location** within the site.
- **escalationId** is set when the submission raised an escalation (low ratings).

Query parameters

Parameter	Type	Required	Description
`dateFrom`	date	Yes	Inclusive start of the date range (ISO-8601 date, `YYYY-MM-DD`). Example: `2026-09-01`.
`dateTo`	date	Yes	Inclusive end of the date range (ISO-8601 date, `YYYY-MM-DD`). Maximum span per request is 92 days. Example: `2026-09-30`.
`orgUnitId`	integer	No	Filter to a single site; omit to return all sites the API key is scoped to. Example: `4021`.
`page`	integer	No	1-indexed page number. Defaults to the first page.
`per_page`	integer	No	Items per page. Default 50, max 500.

Response fields (each item in `data`)

Field	Type	Notes
`feedbackId`	integer	
`orgUnit`	object	
`orgUnit.id`	integer	
`orgUnit.type`	string	One of: `SITE`
`orgUnit.name`	string	e.g. `Head Office - Centurion`
`location`	object, nullable	Area within the site where the QR code / NFC tag / beacon is installed.
`location.id`	integer	
`location.name`	string	e.g. `Level 2 - Ladies Restroom`
`qrCodeId`	string	e.g. `QR-4021-007`
`rating`	integer	1 = very poor, 5 = excellent. Range 1–5.
`category`	string	One of: `CLEANLINESS`, `CONSUMABLES`, `ODOUR`, `SPILL`, `BROKEN_FIXTURE`, `GENERAL`
`comment`	string, nullable	e.g. `Soap dispenser empty.`
`submittedTimestamp`	date-time	When the QR form was submitted.
`escalationId`	integer, nullable	Escalation raised from this submission, if any.

Example request

```
curl -G "$BASE_URL/customer-feedback/qr-responses" \
  -H "X-Api-Key: $FORCELINK_API_KEY" \
  -d dateFrom=2026-09-01 -d dateTo=2026-09-17 -d orgUnitId=4021 -d per_page=1
```

Example response (first item; full envelope in section 3.1)


```
{
  "success": true,
  "data": [
    {
      "feedbackId": 352970,
      "orgUnit": {
        "id": 4021,
        "type": "SITE",
        "name": "Head Office - Centurion"
      },
      "location": {
        "id": 7006,
        "name": "Lift Lobby"
      },
      "qrCodeId": "QR-4021-006",
      "rating": 5,
      "category": "GENERAL",
      "comment": null,
      "submittedTimestamp": "2026-09-01T06:21:00Z",
      "escalationId": null
    }
  ],
  "pagination": {
    "page": 1,
    "per_page": 1,
    "total_items": 491,
    "total_pages": 491
  },
  "meta": "..."
}
```

4.8 Cleaner NFC tag scans and beacon check-ins for a date range, by site

`GET /cleaner-tracking/check-ins`

Returns NFC scans and beacon check-ins whose `checkInTimestamp` falls in the requested range. Check-ins only occur while the cleaner is clocked in (see `/attendance/shift-records`).

- NFC rows are point-in-time scans (**checkOutTimestamp** is null). BEACON rows also carry a check-out time, so dwell time = checkOut – checkIn.
- Check-ins only occur while the cleaner is clocked in.

Query parameters

Parameter	Type	Required	Description
`dateFrom`	date	Yes	Inclusive start of the date range (ISO-8601 date, `YYYY-MM-DD`). Example: `2026-09-01`.
`dateTo`	date	Yes	Inclusive end of the date range (ISO-8601 date, `YYYY-MM-DD`). Maximum span per request is 92 days. Example: `2026-09-30`.
`orgUnitId`	integer	No	Filter to a single site; omit to return all sites the API key is scoped to. Example: `4021`.
`resourceId`	integer	No	Filter to a single cleaner. Example: `88213`.
`page`	integer	No	1-indexed page number. Defaults to the first page.
`per_page`	integer	No	Items per page. Default 50, max 500.

Response fields (each item in `data`)

Field	Type	Notes
`checkInId`	integer	
`resourceId`	integer	Cleaner who checked in (see `/resources`).
`orgUnit`	object	
`orgUnit.id`	integer	
`orgUnit.type`	string	One of: `SITE`
`orgUnit.name`	string	e.g. `Head Office - Centurion`
`location`	object, nullable	Area within the site where the QR code / NFC tag / beacon is installed.
`location.id`	integer	
`location.name`	string	e.g. `Level 2 - Ladies Restroom`
`tagId`	string	NFC tag or beacon identifier.
`method`	string	One of: `NFC`, `BEACON`
`checkInTimestamp`	date-time	NFC scan time, or beacon entry time.
`checkOutTimestamp`	date-time, nullable	Beacon exit time (dwell). Always null for NFC scans.

Example request

```
curl -G "$BASE_URL/cleaner-tracking/check-ins" \
  -H "X-Api-Key: $FORCELINK_API_KEY" \
  -d dateFrom=2026-09-01 -d dateTo=2026-09-17 -d orgUnitId=4022 -d per_page=1
```

Example response (first item; full envelope in section 3.1)


```
{
  "success": true,
  "data": [
    {
      "checkInId": 1145842,
      "resourceId": 88241,
      "orgUnit": {
        "id": 4022,
        "type": "SITE",
        "name": "Parktown Campus - Johannesburg"
      },
      "location": {
        "id": 7015,
        "name": "Ground Floor - Reception"
      },
      "tagId": "NFC-4022-004",
      "method": "NFC",
      "checkInTimestamp": "2026-09-01T06:59:00Z",
      "checkOutTimestamp": null
    }
  ],
  "pagination": {
    "page": 1,
    "per_page": 1,
    "total_items": 1508,
    "total_pages": 1508
  },
  "meta": "..."
}
```

5. How the data fits together

All endpoints share one set of ids, so their results can be joined client-side:

Join	How
Any record → site	`orgUnit.id` (site / org unit). Every record carries the site id and name.
Any person → resource	`employeeId` (attendance), `resourceId` (work orders, check-ins), `raisedByResourceId` / `assignedToResourceId` (escalations) all resolve through `/resources`.
Cleaner → supervisor → manager	`Resource.supervisorId`. Cleaners report to a supervisor and supervisors to the site manager.
QR feedback ↔ escalation	`CustomerFeedbackResponse.escalationId` and `Escalation.feedbackId` point at each other.
Escalation → work order	`Escalation.workOrderId` is the reactive work order raised to deal with it, if one was raised.
Feedback / check-in / escalation → area	`location.id` identifies the area (e.g. Level 2 - Ladies Restroom). A QR code, NFC tag or beacon is fixed to one location.
Stock issue → work order	`StockMovement.referenceDocument` holds the work order code for ISSUE movements.

6. Dashboard recipes

How the dashboard measures can be calculated from the raw data. All of the calculations are done client-side - the API returns raw records only.

Work-order completion time

Source: `/service-proof/completed-work-orders`

completion_hours = (completedDate – createdDate) in hours. Report the median and 90th percentile per site and per workOrderType. Reactive types (Spill Response, Washroom Replenishment, Waste Removal) close within hours; scheduled types can take up to two days.

Customer satisfaction (CSAT)

Source: `/customer-feedback/qr-responses`

Average rating = mean(rating). CSAT % = count(rating ≥ 4) / count(*). Group by orgUnit and location.name to find problem areas, and by category for the low ratings (≤ 2) to see what is going wrong.

Escalation response and resolution

Source: `/escalations`

Time to acknowledge = acknowledgedDate – createdDate; time to resolve = resolvedDate – createdDate. Current backlog = rows with status OPEN or ACKNOWLEDGED. Escalation rate = escalations / QR submissions. Use assignedToResourceId plus /resources to show each supervisor's workload.

Cleaner coverage

Source: `/cleaner-tracking/check-ins + /resources`

Visits per location per day = count of check-ins grouped by location.id and date(checkInTimestamp). Find locations with no visit for more than N minutes during working hours. Beacon dwell = checkOutTimestamp – checkInTimestamp. Roll cleaners up to their supervisor through supervisorId.

Attendance KPIs

Source: `/attendance/shift-records + /resources`

On-time % = ON_TIME / (ON_TIME + LATE). Absenteeism % = ABSENT / all rows. Join to /resources for role and supervisor, and compare against check-ins to confirm that clocked-in cleaners are actually doing rounds.

Stock usage

Source: `/stock/movements + /stock/on-hand`

Consumption = sum of –quantity for ISSUE rows per material per period. Days of cover = quantityOnHand / average daily consumption.

Example - completion time and CSAT by site (Python / pandas):

```
import pandas as pd

wo = pd.json_normalize(list(fetch_all("/service-proof/completed-work-orders",
                                      dateFrom="2026-08-01", dateTo="2026-08-31")))
wo["hours"] = (pd.to_datetime(wo.completedDate)
               - pd.to_datetime(wo.createdDate)).dt.total_seconds() / 3600
print(wo.groupby("orgUnit.name").hours.median())

fb = pd.json_normalize(list(fetch_all("/customer-feedback/qr-responses",
                                      dateFrom="2026-08-01", dateTo="2026-08-31")))
print(fb.groupby("orgUnit.name").rating.agg(avg="mean",
      csat=lambda r: (r >= 4).mean()))
```

7. Known gaps

- **Chi POC areas.** Prestige Pulse is deployed across every site. Areas where Chi is running its POC will be excluded from the QR-feedback and cleaner-tracking data, so that the two dashboards don't show the same data twice. That exclusion is not built yet - every scoped site currently returns full data.
- **SLA flags.** SLA status and breach flags on work orders are out of scope for the POC. Calculate completion time from the timestamps instead.
- **Not real time.** The data is not a live feed. Poll at most every few minutes, and use date ranges rather than re-fetching history.
- **Mock data.** The mock key, site names and values are generated test data, and the generated data ends on 17 Sep 2026. The UAT environment uses the real deployment data.